The Algebraic Bridge: FFGFT and $\mathbb{Z}_3\mathbb{C}$

Frobenius Automorphism, Charge Classification and
Five Structural Questions between $T^4/\mathbb{Z}_3$ and GF(9)

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Abstract

This document investigates five algebraic structural questions between FFGFT geometry ($T^4/\mathbb{Z}_3$, Docs. 257/321) and Matzke's GALG framework (G(6,$\mathbb{Z}_3\mathbb{C}$)). Three main results **[B]**: (1) The FFGFT sector pairing $k \leftrightarrow -k$ and Matzke's $V_s = V^\dagger$ both realise the Frobenius automorphism $x \mapsto x^3$ in GF(9). (2) The canonical ring homomorphism $\varphi : N \mapsto N \bmod 3$ connects Furey's number operator with the FFGFT winding quantum. (3) The injection $\iota : Q \mapsto (3Q \bmod 3, \lfloor Q \rfloor)$ shows that $N \bmod 3$ fully classifies the charge $Q = N/3$.

Verification scripts: `pruef_324_z3c_trine_exakt.py`, `pruef_325_gf9_ffgft_bruecke.py`.

Notation: Symmetric $\mathbb{Z}_3\mathbb{C}$ after Matzke: elements $a + bi$ with $a, b \in \{-1, 0, +1\}$, where $+1 + 1 = -1$ (ternary inline carry).

.1 Motivation: Characteristic 3 and FFGFT

In characteristic 3, $(1+N)^3 = 1$ for nilpotent N with $N^2 = 0$ holds *algebraically exactly* — because $\binom{3}{k} \equiv 0 \pmod 3$ for $0 < k < 3$. The machine-epsilon result $|T_k^3 - \mathbf{1}| < 6.5 \times 10^{-16}$ from Doc. 324 is therefore not an approximation but a consequence of an algebraic theorem **[B]**.

.2 Five structural questions — updated status

Question 1: Number operator N and FFGFT winding quantum Δw

Previously open [[S]]; now partially closed.

Furey's number operator $N = \sum_j q_j^\dagger q_j$ has spectrum $\{0, 1, 1, 1, 2, 2, 2, 3\}$ on the eight basis states of one generation. Reduction mod 3 gives:

$$N \bmod 3 = \{0, +1, +1, +1, -1, -1, -1, 0\} \tag{1}$$

with particle classification:

$N \bmod 3 = 0$: colour-neutral (neutrino, positron)
$N \bmod 3 = 1$: anti-down quarks (3 colours)
$N \bmod 3 = 2$: up quarks (3 colours)

This is *exactly* the $\mathbb{Z}_3$ class decomposition of the FFGFT sectors $\Delta w \in \{0, +1, -1\}$ on $\mathrm{T}^4/\mathbb{Z}_3$.

Ring homomorphism [B]:

$$\varphi : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}, \quad \varphi(N) = N \bmod 3 = \Delta w \tag{2}$$

is the canonical quotient; $\varphi(N+M) = \varphi(N) + \varphi(M) \bmod 3$ is a proved algebraic property.

Status: [B] (homomorphism), [K] (physical assignment)

$\varphi(N) = N \bmod 3$ is a ring homomorphism **[B]**. The physical identification $\varphi(N) = \Delta w$ (Furey's number operator ↔ FFGFT winding quantum) classifies the same threefold particle spectrum decomposition; the deeper algebraic reason for this agreement is **[K]**.

Question 2: Charge formula $Q = N/3$ and $\mathbb{Z}_3$ classification

Previously open [[S]]; now partially closed.

$Q = N/3 \in \mathbb{Q}$ cannot be embedded into GF(3) (3 is a zero divisor in $\mathbb{Z}/9\mathbb{Z}$, no inverse). However an injection exists:

$$\iota : \{0, \tfrac{1}{3}, \tfrac{2}{3}, 1\} \hookrightarrow \mathbb{Z}/3\mathbb{Z} \times \{0,1\}, \quad \iota(Q) = (3Q \bmod 3,\ \lfloor Q \rfloor) \tag{3}$$

with values:

Q	$\iota(Q)$	Particle
0	$(0,0)$	Neutrino
$1/3$	$(1,0)$	Anti-Down
$2/3$	$(2,0)$	Up
1	$(0,1)$	Positron

The $\mathbb{Z}/3\mathbb{Z}$ component of $\iota(Q)$ is exactly $N \bmod 3$.

Status: [B] (classification), [K] (Q as $\mathbb{Q}$ number)

N mod 3 fully classifies $Q = N/3$ **[B]**. Q itself (as rational number $1/3$, $2/3$) requires $\mathbb{Q}$ and is not a GF(3) object **[K]**.

Question 3: Sector pairing $k \leftrightarrow -k$ and Frobenius automorphism

Main result; was [S]; now [B].
The Frobenius automorphism in GF(9):

$$\phi : x \mapsto x^3, \quad \phi(a+bi) = a - bi \quad \text{[field conjugation]} \tag{4}$$

operates simultaneously on three levels:

- **GF(9) field:** $a + bi \mapsto a - bi$ (conjugation)
- **FFGFT sector:** $k \mapsto -k$ mod 3 ($\mathbb{Z}_3$ inversion)
- **Matzke vacuum:** $V \mapsto V_s = V^\dagger$ (matrix adjunction)
- `pruef_326_gf9_orbit_masse_weinberg.py`: Fixed points **[B]**, norm hierarchy **[B]**, Weinberg topology **[K]**

Numerically verified: $x^3 = \bar{x}$ for all 8 non-zero GF(9) elements **[B]**.

Status [B]

All three operations are algebraically isomorphic realisations of the Frobenius automorphism $x \mapsto x^3$.

Question 4: Casimir singularity and ξ in GF(3)

$\xi = (4/3)/10^4$: Casimir denominator $3 \equiv 0$ in GF(3) is singular. ξ mod $3 = 1$ — ξ lies in residue class 1. The singularity marks the transition GF(3)$\to \mathbb{R}$ **[K]**.

Question 5: $\sqrt{2}$ in GF(9) and n_{thresh}

$i^2 = -1 \equiv 2 \pmod 3$, so $\sqrt{2} = i \in \text{GF}(9)$ **[B]**. $n_{\text{thresh}} = 2\pi/(\sqrt{2}\ln 2) \approx 6.41$ (R93) thus has an algebraically exact GF(9) kernel.

.3 Three further structural results

Frobenius fixed points = FFGFT sectors (Question 3, extended)

The complete orbit structure under $\phi(x) = x^3$ in GF(9):

- **Fixed points** ($x^3 = x$): $\{0, +1, -1\} = \text{GF}(3)$ — exactly the FFGFT sectors $\Delta w \in \{0, +1, -1\}$ **[B]**
- **Non-fixed points**: 6 elements in 3 conjugate 2-cycles **[B]**: $(i \leftrightarrow 2i)$, $(1{+}i \leftrightarrow 1{+}2i)$, $(2{+}i \leftrightarrow 2{+}2i)$ — exactly the FFGFT pairs $(k, -k)$ for $k \neq 0$.

Stable states under Frobenius are the sectors; unstable states are the conjugate pairs.

Norm hierarchy GF(3) → GF(9) → $\mathbb{R}(\xi)$ for masses

In GF(3), $1^2 \equiv 2^2 \equiv 1 \pmod 3$ — muon and tau are *algebraically degenerate* as squares **[B]**. The GF(3) structure distinguishes only:

$$n^2 \in \{0,\ 1\} \pmod 3: \quad 0 \leftrightarrow \text{massless}, \quad 1 \leftrightarrow \text{massive (generations 2+3 degenerate)} \tag{5}$$

In GF(9), the norm $\|x\| = (a^2 + b^2) \bmod 3$ gives two classes **[B]**:

$$\begin{aligned} \|x\| = 1 &: \{+i, -i, +1, -1\} && \text{— purely real/imaginary (light)} \\ \|x\| = -1 &: \{+1+i, +1-i, -1+i, -1-i\} && \text{— mixed (heavy)} \end{aligned}$$

The full mass hierarchy requires ξ **[K]**: ξ breaks the GF(3) degeneracy and gives $m_\mu/m_e \approx 207$, $m_\tau/m_\mu \approx 17$ as continuous numbers.

Status: [B] (degeneracy), [K] (mass values)

Hierarchy: GF(3) gives whether massive/massless; GF(9) gives light/heavy; $\xi \in \mathbb{R}$ gives exact mass ratios. The singularity $C_2 = 4/3$ in GF(3) (Question 4) marks exactly the transition GF(3)→ $\mathbb{R}$.

Weinberg angle: GF(9) gives topology, ξ gives value

Gal(GF(9)/GF(3)) $= \mathbb{Z}_2$ explains *why* there are two gauge groups SU(2)$_L$ and U(1)$_Y$ — a degree-2 extension gives exactly two components **[K]**.

The best GF(9) candidate for $\sin^2\theta_W$ is $|\mathrm{GF}(3)^*|/|\mathrm{GF}(9)^*| = 2/8 = 0.250$, compared to the experimental value 0.231 (8% deviation). The exact value is reached by ξ corrections (Doc. 323, **[K]**).

Status [K]

GF(9) provides the algebraic justification for two gauge groups (degree-2 Galois extension) and a tree-level approximation ≈ 0.25. The numerical value $\sin^2\theta_W = 0.231$ comes from ξ.

.4 Status overview

Verification scripts

- `pruef_324_z3c_trine_exakt.py`: Algebraic proof $T_k^3 = \mathbf{1}$ in char. 3 **[B]**
- `pruef_325_gf9_ffgft_bruecke.py`: All five questions; Frobenius **[B]**; ring homo **[B]**; injection ι **[B]**

Table 1: Algebraic bridge questions FFGFT ↔ GF(9) — updated status

#	Question	Result	Status
1a	Ring homo $\varphi(N) = N$ mod 3	canonical, proved	**[B]**
1b	$\varphi(N) = \Delta w$ physical	same threefold split	**[K]**
2a	N mod 3 classifies Q	injection ι proved	**[B]**
2b	$Q = N/3$ as GF(3) object	impossible ($3 \equiv 0$)	**[K]**
3	$k \leftrightarrow -k$ vs. $V_s = V^\dagger$	Frobenius $x \mapsto x^3$	**[B]**
4	$C_2 = 4/3$ and ξ in GF(3)	singularity = GF(3)→ $\mathbb{R}$	**[K]**
5	$\sqrt{2}$ in GF(9), n_{thresh}	$i^2 = 2$, kernel exact	**[B]**
P3b	Fixed pts. = sectors, $(k, -k)$ pairs	orbit structure	**[B]**
P4b	$1^2 = 2^2 = 1$ in GF(3)	Gen. 2+3 degenerate	**[B]**
P4c	Norm $\in \{1, 2\}$ in GF(9)	light/heavy	**[B]**
P5b	$\lvert\mathrm{GF}(3)^*\rvert / \lvert\mathrm{GF}(9)^*\rvert = 1/4$	tree-level $\sin^2\theta_W$	**[K]**